



Worksheet 2

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Semester: 2nd

Subject Name: DBMS LAB

UID: 25MCI10082

Section/Group: 1/A

Date of Performance: 30/03/2026

Subject Code:

1. Aim of the Session

Implementation of joins in PostgreSQL (inner join ,left join,right join, self join and cross join).

2. Objective of the Session

Apply joins to a real-world database schema (e.g., Students, Courses, Enrollments, Departments)

3. Practical / Experiment Steps

Design the database schema for different tables.

Create table using appropriate constraints.

Insert sample records into tables.

Perform joins.

4. Procedure of the Practical

(i) Start the system and log in to the computer.

(ii) Open PostgreSQL software.

(iii) create database Experiment7;



(iv) Create a tables.

CREATE TABLE departments (

department_id INT PRIMARY KEY,

department_name VARCHAR(50)

);

CREATE TABLE students (

student_id INT PRIMARY KEY,

name VARCHAR(50),

department_id INT,

FOREIGN KEY (department_id) REFERENCES departments(department_id)

);

CREATE TABLE courses (

course_id INT PRIMARY KEY,

course_name VARCHAR(50)

);

CREATE TABLE enrollments (

student_id INT,

course_id INT,

PRIMARY KEY (student_id, course_id),

FOREIGN KEY (student_id) REFERENCES students(student_id),

FOREIGN KEY (course_id) REFERENCES courses(course_id)

);



(v) Insert records using DML commands.

INSERT INTO departments VALUES

(1, 'Computer Science'),

(2, 'Mechanical'),

(3, 'Electrical');

INSERT INTO students VALUES

(1, 'Aryan', 1),

(2, 'Rohit', 2),

(3, 'Priya', 1),

(4, 'Simran', 3),

(5, 'Karan', NULL);

INSERT INTO courses VALUES

(101, 'DBMS'),

(102, 'OS'),

(103, 'Maths'),

(104, 'Physics');

INSERT INTO enrollments VALUES

(1, 101),

(1, 102),

(2, 103),

(3, 101);

(vi) Queries to list students with their enrolled courses (INNER JOIN).

```
SELECT s.name,c.course_name  
  
FROM students s  
  
INNER JOIN enrollments e ON s.student_id = e.student_id  
  
INNER JOIN courses c ON e.course_id = c.course_id;
```

	name character varying (50) 🔒	course_name character varying (50) 🔒
1	Aryan	DBMS
2	Aryan	OS
3	Rohit	Maths
4	Priya	DBMS

(vii) Students not enrolled in any course (LEFT JOIN).

```
SELECT s.name  
  
FROM students s  
  
LEFT JOIN enrollments e ON s.student_id = e.student_id  
  
WHERE e.course_id IS NULL;
```

	name character varying (50) 🔒
1	Karan
2	Simran

(viii) Courses with or without enrolled students (RIGHT JOIN).

```
SELECT c.course_name, s.name  
  
FROM students s  
  
RIGHT JOIN enrollments e ON s.student_id = e.student_id  
  
RIGHT JOIN courses c ON e.course_id = c.course_id;
```

	course_name character varying (50) 🔒	name character varying (50) 🔒
1	DBMS	Aryan
2	OS	Aryan
3	Maths	Rohit
4	DBMS	Priya
5	Physics	[null]

(ix) Students with department info using SELF JOIN or multiple joins.

```
SELECT s.name, d.department_name  
  
FROM students s  
  
LEFT JOIN departments d ON s.department_id = d.department_id;
```

	name character varying (50) 🔒	department_name character varying (50) 🔒
1	Aryan	Computer Science
2	Rohit	Mechanical
3	Priya	Computer Science
4	Simran	Electrical
5	Karan	[null]

(x) Display all possible student-course combinations (CROSS JOIN).

```
SELECT s.name, c.course_name
```

```
FROM students s
```

```
CROSS JOIN courses c;
```

	name character varying (50) 	course_name character varying (50) 
1	Aryan	DBMS
2	Rohit	DBMS
3	Priya	DBMS
4	Simran	DBMS
5	Karan	DBMS
6	Aryan	OS
7	Rohit	OS
8	Priya	OS
9	Simran	OS
10	Karan	OS
11	Aryan	Maths
12	Rohit	Maths
13	Priya	Maths
14	Simran	Maths
15	Karan	Maths
16	Aryan	Physics
17	Rohit	Physics
18	Priya	Physics
19	Simran	Physics
20	Karan	Physics

Learning Outcomes

- Understand the concept of relational databases and how data is distributed across multiple tables.
- Apply different types of SQL joins such as INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and CROSS JOIN.
- Retrieve meaningful information by combining data from multiple related tables like Students, Courses, Enrollments, and Departments.
- Identify the appropriate join type based on the problem requirement and data relationship.
- Analyze query results to understand how NULL values and unmatched records are handled in different joins.